P08

Coupled Resonator Arrays and Bandpass Transmission Lines

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Domain DOM_ENG

Abstract

Chains of inductively coupled LC resonators support bandpass electromagnetic propagation. Natural modes correspond to standing wave eigenvectors identical to acoustic phonon dispersion in crystal lattices.

Abstract quoted from G. L. Matthaei et al., Microwave Filters and Coupling Structures (1964).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?